

ISEN/DAEN 427 Lecture 5

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Bayesian Inference with Beta Prior and Binomial Likelihood Cue Combination

BM Ch. 5, 6



Bayesian Inference with Binomial Likelihood

- Assume observations $x_1, \dots, x_N$ are binary $x_i \in \{0, 1\}$ and $s \in [0, 1]$.
- Given a likelihood $\mathbb{P}(x_i|s)$, conditional independence implies:

$$\begin{aligned}\mathbb{P}(x_1, \dots, x_N | s) &= \prod_{i=1}^N \mathbb{P}(x_i | s) \\ &= \prod_{i=1}^N s^{x_i} (1 - s)^{1-x_i} \\ &= s^{\sum_{i=1}^N x_i} (1 - s)^{N - \sum_{i=1}^N x_i}\end{aligned}$$

where $\sum_{i=1}^N x_i$ is the total number of ones ("successes") and $N - \sum_{i=1}^N x_i$ are the total number of zeros ("failures").

Bayesian Inference with Binomial Likelihood

- The likelihood can alternatively be written in terms of the Binomial distribution for $y = \sum_{i=1}^N x_i$:

$$\mathbb{P}(y|s) = \binom{N}{y} s^y (1-s)^{N-y}$$

where $y \in \{0, 1, \dots, N\}$.

- Recall that

$$\mathbb{E}[y] = Ns \quad \text{Var}[y] = Ns(1-s)$$

Beta vs Binomial in Bayesian Inference

- **Binomial distribution** models **data**:
 - Number of successes in n trials given probability p .
 - Example: How many heads in 10 coin flips?
- **Beta distribution** models **beliefs about** p :
 - Represents uncertainty over the unknown probability itself.
 - Example: How likely is $p = 0.7$?

In Bayesian inference, we often want to infer p from data.
The Beta distribution is ideal for this.

Conjugacy: Beta is conjugate prior of the Binomial.

$$p \sim \text{Beta}(\alpha, \beta), \quad \text{data} \sim \text{Binomial}(n, p)$$

$$p \mid \text{data} \sim \text{Beta}(\alpha + \text{successes}, \beta + \text{failures})$$

- Posterior is also Beta $\Rightarrow$ simple closed-form updates.
- Flexible: α and β can express uniform, peaked, or skewed beliefs.
- Binomial is used as **likelihood**, Beta is used as **prior and posterior**.

Use Binomial for data, Beta for beliefs.
Easily update with new evidence.

Note: *What is Conjugacy?*

- In Bayesian inference:

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

- A prior and likelihood are **conjugate** if the posterior has the **same distribution family** as the prior.

Example: Beta–Binomial Conjugacy

$$p \sim \text{Beta}(\alpha, \beta), \quad \text{Data} \sim \text{Binomial}(n, p)$$

$$p \mid \text{data} \sim \text{Beta}(\alpha + \text{successes}, \beta + \text{failures})$$

- Posterior stays in the Beta family $\rightarrow$ simple math.
- Easy to update beliefs as new data arrives.
- No complicated integrals needed.

Prior: Beta distribution

- The beta distribution with parameters $\alpha > 0$ and $\beta > 0$ is of the form:

$$\mathbb{P}(s) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} s^{\alpha-1} (1 - s)^{\beta-1}$$

- The mean and variance: $\mathbb{E}[s] = \frac{\alpha}{\alpha + \beta}$ and $\text{Var}[s] = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$

The Beta distribution is a type of probability distribution.

- It describes probabilities for values between 0 and 1 (like percentages).
- It is controlled by two numbers: α and β .

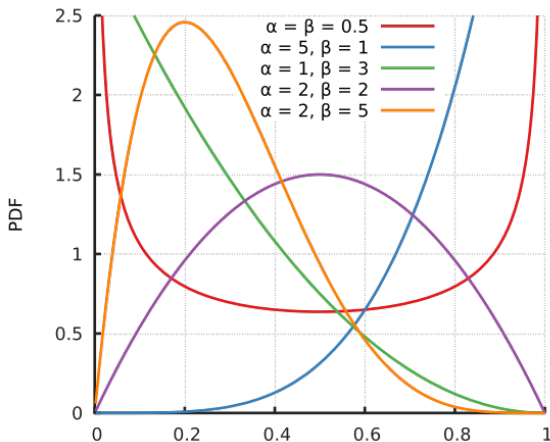
The Beta distribution is great for modeling situations like:

- The probability of success in experiments (e.g., shooting hoops).
- Estimating proportions when you have some data but not too much.
- Bayesian statistics (used to update beliefs as new data comes in).

Different Shapes

Depending on α and β , the shape changes:

- $\alpha = \beta = 1 \rightarrow$ Flat (every value equally likely).
- $\alpha > \beta \rightarrow$ Skewed toward 1 (more likely high values).
- $\alpha < \beta \rightarrow$ Skewed toward 0 (more likely low values).



Learning as Inference

Assume the a priori distribution is **beta** with parameters $\alpha > 0$ and $\beta > 0$ and $y = \sum_{i=1}^N x_i$ then

$$\begin{aligned}\mathbb{P}(s|y) &= \kappa \mathbb{P}(y|s) \mathbb{P}(s) \\&= \kappa \binom{N}{y} s^y (1-s)^{N-y} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} s^{\alpha-1} (1-s)^{\beta-1} \\&= \kappa \binom{N}{y} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} s^{y+\alpha-1} (1-s)^{N+\beta-y-1}\end{aligned}$$

where

$$\kappa = \frac{1}{\int_0^1 \mathbb{P}(y|s) \mathbb{P}(s) ds} = \frac{1}{\mathbb{P}(y)}$$

Computing the Posterior

- After recording y ones ("successes") out of $N > 1$ the posterior distribution is **beta** with parameters α' and β' , i.e.:

$$\mathbb{P}(s|y) = \frac{\Gamma(\alpha' + \beta')}{\Gamma(\alpha')\Gamma(\beta')} s^{\alpha'-1} (1-s)^{\beta'-1}$$

- Thus,

$$\mu_{post} = \frac{\alpha'}{\alpha' + \beta'} = \frac{y + \alpha}{N + \alpha + \beta}$$

$$\begin{aligned}\sigma_{post} &= \frac{\alpha' \beta'}{(\alpha' + \beta')^2 (\alpha' + \beta' + 1)} \\ &= \frac{(y + \alpha)(N - y + \beta)}{(N + \alpha + \beta)^2 (N + \alpha + \beta + 1)}\end{aligned}$$

Computing the Posterior

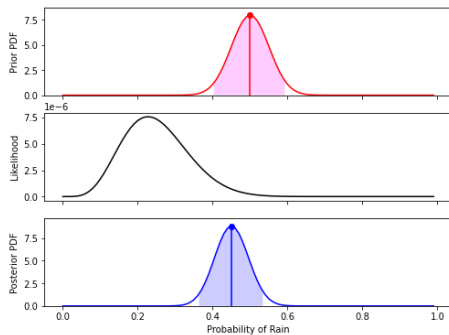


Figure 1: Posterior with Beta Prior and Binomial Likelihood
 $s = \mathbb{P}(\text{"rain"})$

Connection to Reinforcement Learning

- With $\alpha = \beta = 1$, an additional $N + 1$ observation leads to the posterior mean

$$\mu_{post,N+1} = \frac{y + x_{N+1} + 1}{N + 1 + 2}$$

- A recursive update equation for the posterior mean

$$\begin{aligned}\mu_{post,N+1} &= \frac{y + 1}{N + 2} \frac{N + 2}{N + 3} + x_{N+1} \frac{1}{N + 3} \\ &= \mu_{post,N} \left(1 - \frac{1}{N + 3}\right) + x_{N+1} \frac{1}{N + 3} \\ &= \mu_{post,N} + \frac{1}{N + 3} (x_{N+1} - \mu_{post,N})\end{aligned}$$

Connection to Reinforcement Learning

- This equation resembles the *Rescorla-Wagner rule* in reinforcement learning.
- New information or “surprise” is $x_{N+1} - \mu_{post,N}$.
- The factor $\frac{1}{N+3}$ acts as the learning rate.

Learning happens through the difference between
expected and **actual** outcomes.

- If the outcome is **better than expected**, the prediction increases.
- If the outcome is **worse than expected**, the prediction decreases.
- Over time, the agent learns to make more accurate predictions.

Rescorla–Wagner Rule: Formula

$$V_{t+1} = V_t + \alpha(\lambda - V_t)$$

- V_t = current prediction (expected reward)
- λ = actual outcome
- α = learning rate ($0 < \alpha \leq 1$)
- $\lambda - V_t$ = **prediction error**

Example: If a student expects 60% but scores 80%, the prediction error is +20%, leading to an upward update.

Cue Combination

- Inference based on two *different* types of measurements (cues).
- How should cues be processed in order to optimize inference?
- Experimental evidence indicates the human brain **combines** cues.

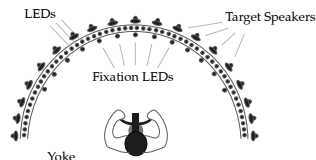


Figure 2: Testing Visual and Auditory Cue Combination

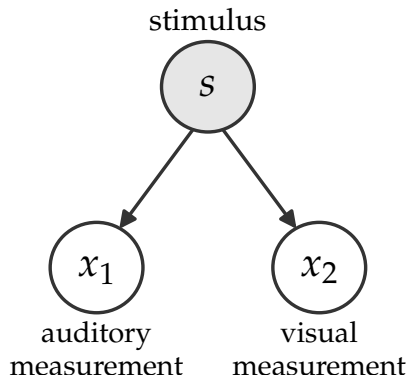


Figure 3: Generative Model

- Let x_1 and x_2 denote the observation values (cues). We assume

$$\mathbb{P}(x_1, x_2 | s) = \mathbb{P}(x_1 | s) \mathbb{P}(x_2 | s)$$

- Let us derive the posterior when

$$\begin{aligned} \mathbb{P}(x_i | s) \\ = \\ \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{1}{2\sigma_i^2}(x_i - s)^2\right) \end{aligned}$$

for $i = 1, 2$.

Posterior Distribution

- Similarly to derivation in Lecture 4, it can be shown that:

$$\mathbb{P}(s|x_1, x_2) = \frac{1}{\sqrt{2\pi}\sigma_{post}} \exp\left(-\frac{1}{2\sigma_i^2}(s - \mu_{post})^2\right)$$

with

$$\mu_{post} = \frac{\frac{1}{\sigma_s^2}}{\frac{1}{\sigma_s^2} + \frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}} \mu_s + \frac{\frac{1}{\sigma_1^2}}{\frac{1}{\sigma_s^2} + \frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}} x_1 + \frac{\frac{1}{\sigma_2^2}}{\frac{1}{\sigma_s^2} + \frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}} x_2$$

$$\sigma_{post} = \frac{1}{\frac{1}{\sigma_s^2} + \frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}}$$

Posterior Distribution

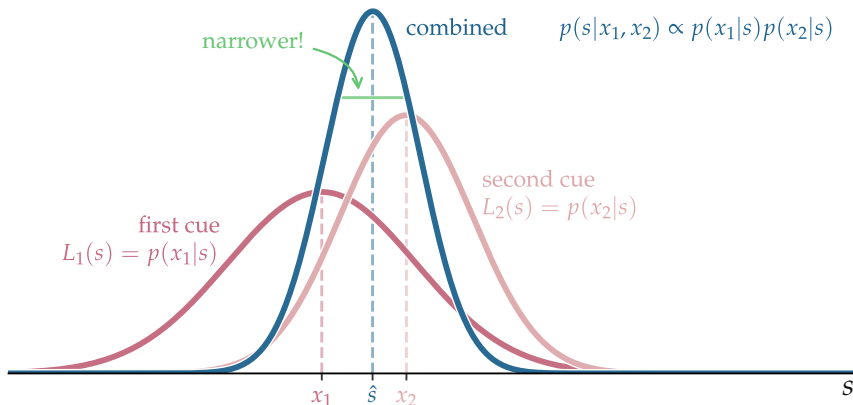


Figure 4: Posterior Distribution from Cue Combination